

EDMONTON NAMAQ, AB, Canada

WMO: 711210

Lat: 53.6639N

Lon: 113.4661W

Elev: 688

StdP: 93.33

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.8	(c) -26.7	(d) -34.9	(e) 0.1	(f) -29.6	(g) -31.7	(h) 0.2	(i) -26.5	(j) 11.1	(k) -5.6	(l) 9.9	(m) -5.8	(n) 2.6	(o) 270	(p) 0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	27.8	17.7	26.1	16.9	24.4	16.0	19.1	25.7	18.0	24.4	17.0	22.9	4.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.8	22.4	15.4	11.9	20.9	14.4	11.1	19.8	57.2	25.8	53.5	24.4	50.2	23.0	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.1	(b) 8.7	(c) 7.6	(e) -34.2	(f) 30.9	(g) 3.6	(h) 1.8	(i) -36.8	(j) 32.2	(k) -38.9	(l) 33.2	(m) -40.9	(n) 34.2	(o) -43.6	(p) 35.6		
(a) 10.1	(b) 8.7	(c) 7.6	(e) -34.3	(f) 21.3	(g) 3.5	(h) 1.3	(i) -36.8	(j) 22.3	(k) -38.9	(l) 23.0	(m) -40.8	(n) 23.7	(o) -43.3	(p) 24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-10.4	-10.3	-4.4	3.7	10.5	14.5	17.0	15.7	11.2	4.1	-4.1	-10.3
	DBStd	11.81	8.61	8.03	7.68	5.66	4.60	3.13	2.99	3.17	4.21	4.85	6.68	7.83
	HDD10.0	3174	633	567	445	199	49	3	1	2	35	188	422	629
	HDD18.3	5571	892	801	703	441	244	120	62	92	217	440	672	887
	CDD10.0	684	0	0	0	9	65	139	217	178	70	7	0	0
	CDD18.3	39	0	0	0	0	2	6	20	10	1	0	0	0
	CDH23.3	597	0	0	0	2	48	91	244	157	53	2	0	0
	CDH26.7	105	0	0	0	0	6	14	52	24	9	0	0	0
Wind	WSAvg	3.5	3.4	3.4	3.6	4.1	4.2	3.6	3.3	3.0	3.4	3.6	3.4	3.3
Precipitation	PrecAvg	420	18	12	16	27	49	69	90	54	27	20	15	17
	PrecMax	554	84	30	47	47	100	146	145	114	77	54	34	34
	PrecMin	235	3	5	5	3	8	20	19	17	1	4	3	5
	PrecStd	80	17	6	9	13	24	31	31	25	19	13	8	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	7.5	12.6	22.7	27.4	28.3	30.9	29.1	28.0	21.7	12.5	7.1
		MCWB	2.4	3.0	5.9	11.2	14.4	17.2	20.0	18.1	16.3	12.5	6.6	2.9
	2%	DB	4.9	4.6	9.2	19.0	24.5	25.9	27.9	27.0	24.6	16.8	8.5	4.3
		MCWB	1.5	0.8	3.5	9.1	13.0	15.8	18.9	17.8	15.0	9.4	4.1	0.9
	5%	DB	3.2	2.7	6.8	15.8	22.2	23.7	26.1	25.1	22.0	14.0	6.1	2.4
		MCWB	0.3	-0.3	2.4	7.4	11.8	14.9	17.9	17.3	14.0	7.8	2.6	-0.4
	10%	DB	1.2	0.8	4.8	12.9	19.6	21.8	24.1	23.2	19.2	11.6	4.1	0.4
		MCWB	-1.2	-1.6	1.2	5.9	10.6	13.9	16.9	16.3	12.4	6.3	1.3	-1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.2	6.1	11.4	15.3	18.6	21.2	20.0	17.2	12.7	6.7	3.1
		MCDB	6.1	7.3	11.9	21.5	25.3	26.1	27.6	25.9	25.8	20.6	11.3	6.9
	2%	WB	1.7	1.1	3.9	9.6	13.7	16.8	19.7	18.6	15.6	10.0	4.5	1.1
		MCDB	4.6	4.1	8.4	18.2	22.3	23.8	26.3	25.2	23.6	15.9	8.0	3.8
	5%	WB	0.4	-0.1	2.5	7.9	12.4	15.7	18.5	17.6	14.3	8.2	2.9	-0.3
		MCDB	3.1	2.5	6.4	15.1	21.0	22.2	24.9	24.0	21.1	12.9	5.5	2.4
	10%	WB	-1.2	-1.6	1.2	6.3	11.1	14.6	17.4	16.6	12.9	6.9	1.5	-2.0
		MCDB	1.1	0.9	4.5	12.3	18.5	20.4	23.1	22.4	18.3	10.9	3.8	0.4
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.6	9.6	11.1	12.5	11.3	11.3	12.1	12.2	10.5	8.4	8.6
		MCDBR	9.5	11.1	11.5	16.3	16.3	14.6	14.6	15.5	17.0	14.7	10.8	9.6
		MCWBR	7.3	8.4	7.7	9.1	8.0	7.0	7.3	7.7	8.8	8.6	7.3	7.4
	5% WB	MCDBR	9.4	10.7	11.0	15.6	14.9	12.9	13.5	14.2	16.0	13.9	10.0	9.6
		MCWBR	7.5	8.3	7.6	9.1	7.6	6.6	7.2	7.5	8.5	8.4	7.2	7.7
Clear-Sky Solar Irradiance	taub	0.256	0.266	0.311	0.325	0.341	0.352	0.354	0.361	0.307	0.281	0.253	0.239	
	taud	2.307	2.354	2.303	2.391	2.395	2.381	2.419	2.352	2.502	2.549	2.456	2.315	
	Ebn at Noon	732	851	872	907	906	897	888	858	876	833	749	684	
	Edh at Noon	56	75	99	104	110	113	107	108	81	62	49	46	
All-Sky Solar Radiation	RadAvg	0.96	2.00	3.54	4.67	5.72	6.02	6.15	5.06	3.49	2.00	0.95	0.71	
	RadStd	0.11	0.15	0.27	0.44	0.43	0.39	0.30	0.35	0.36	0.24	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page